Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 11 [2024 P2]

Test Date: 11th May 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2024-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with stem NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with stem NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 37)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 38 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with stem NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

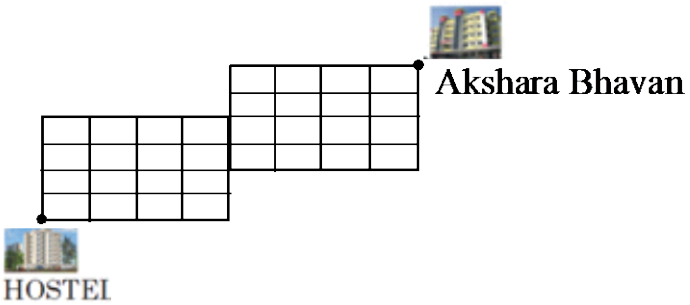
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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1. Let m_1, m_2 be the slopes of two adjacent sides of a square of side a such that $a^2 + 11a + 3\left(m_1^2 + m_2^2\right) = 220$. If one vertex of the square is $(10(\cos\alpha - \sin\alpha), 10(\sin\alpha + \cos\alpha))$, where $\alpha \in \left(0, \frac{\pi}{2}\right)$ and the equation of one diagonal is $(\cos\alpha - \sin\alpha)x + (\sin\alpha + \cos\alpha)y = 10$, then $72(\sin^4\alpha + \cos^4\alpha) + a^2 - 3a + 13$ is equal to

A) 119 **B)** 128 **C)** 145 **D)** 155

2. Number of shortest ways (consider taking one step at a time along right side or upward direction among junctions) from HOSTEL to Akshara Bhavan is equal to (as shown in given figure)



A) 2275 **B)** 2575 **C)** 2725 **D)** 2250

3. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that $f(x+1) = \frac{1}{2} f(x)$ for all $x \in \mathbb{R}$, and let $a_n = \int_0^n f(x)\,dx$ for all integers $n \geq 1$. Then, which of the following is/are CORRECT?

A) $\lim_{n\to\infty} a_n$ exists and equals $\int_0^1 f(x)\,dx$

B) $\lim_{n\to\infty} a_n$ does not exist

C) $\lim_{n\to\infty} a_n$ exists if and only if $\left|\int_0^1 f(x)dx\right| < 1$

D) $\lim_{n\to\infty} a_n$ exists and equals $2\int_0^1 f(x)dx$

4. Let $a = 8$ and $b = 3^9$ and we define a sequence $\{\mu_n\}$ as follows:

$$\mu_1 = b, \mu_{n+1} = \begin{cases} \frac{1}{3}\mu_n, if\ \mu_n\ is\ multiple\ of\ 3 \\ \mu_n + a, otherwise \end{cases}$$ then $\mu_{500} - \mu_{300} - \mu_{400} =$

A) 1 **B)** 3 **C)** 5 **D)** 7

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Number of ways in which 11 letters: A, B, C, D, E, F, X, X, X, Y, Y can be arranged

A) in a row so that every 'Y' lies between two X's (not necessarily adjacent) and one X lies between two Y's (not necessarily adjacent) equals ${}^{11}P_5$

B) in a row so that every 'Y' lies between two X's (not necessarily adjacent) and one X lies between two Y's (not necessarily adjacent) equals ${}^{11}P_6$

C) in a row so that A, B, C, D, E, F are arranged in alphabetic order (not necessarily adjacent) equals 4620

D) in a row so that A, B, C, D, E, F are arranged in alphabetic order (not necessarily adjacent) equals 2310

6. Let $P(n) = P_n = \sum_{r=0}^{n} \frac{(-1)^r r}{r+1} {}^nC_r$. Now which of the following holds good?

A) $|P_{10}|$ is harmonic mean of $|P_9|$ & $|P_{11}|$

B) $\sum_{r=5}^{10} P(r)P(r-1) = -\frac{6}{55}$

C) $|P_{10}|$ is arithmetic mean of $|P_9|$ & $|P_{11}|$

D) $\sum_{r=5}^{10} P(r)P(r-1) = \frac{6}{55}$

7. A tangent is drawn at any point P(t) on the parabola $y^2 = 8x$ and a point $Q(\alpha, \beta)$ is taken on it from which pair of tangents QA and QB are drawn to the circle $x^2 + y^2 = 4$, where A & B are points of contact. Now if point P(t) varies on the parabola then the locus of all such point of concurrency of the chord of contact AB is denoted by S = 0, then which of the following is/are CORRECT?

A) Equation of S = 0 is $2x + y^2 = 0$

B) Equation of S = 0 is $x + y^2 = 0$

C) Area bounded by S = 0 and the line $2x + 1 = 0$ is $\frac{2}{3}$ sq. units

D) Tangents drawn at the extremities of intersection of S = 0 and the line 2x – y +1 = 0 to the curve S = 0 includes an angle 75°

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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8. Consider the following frequency distribution

X	A	2A	3A	4A	5A	6A
f	2	1	1	1	1	1

Where, A is a positive integer and has variance 160. Then the value of A is

9. The number of elements in the set $\{\frac{a}{b} \in I^+ : 2a^2 + 3b^2 = 35, a, b \in Z\}$, where Z is the set of all integers, is

10. Assume that $f(1) = 0$ and that for all integers m and n, $f(m+n) = f(m) + f(n) + 3(4mn - 1)$, then unit(s) digit of the value $f(19)$ is

11. Volume of parallelepiped formed by planes $2x+2y+z=4$, $2x+2y+z=1$, $2x-y+z=3$, $2x-y+z=6$, $3x+4y+12z=-1$, $3x+4y+12z=12$ is V, then the value of $\frac{14V}{13}$ is

12. Let 'f' be continuous periodic and bounded function with period 3 such that $\int_0^3 f(t)dt=6$.

Also, $F(x)=\int_0^x f(t)dt$. The value of $\lim_{x\to 0} xF\left(\frac{1}{x}\right)$ is equal to

13. Let $S=\left\{\theta:7\cos^2\theta-3\sin^2\theta-2\cos^2 2\theta=2,\theta\in(0,2\pi)\right\}$. Let S_1 be the set of monic quadratic equation(s) in the form of $x^2-2\left(\tan^2\theta+\cot^2\theta\right)x+6\sin^2\theta=0\ \forall\theta\in S$. Let S_2 the set of roots of all the equations given in S_1. sum of element(s) of S_2 is ____

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

If the value(s) of A for which the equation $\cot^3 A+\cot^2 A\left|\cot A+x\right|+\left|x\cot^2 A+1\right|=1$ has not less than 6 different solutions which are integers are

$\left[\cot^{-1}\alpha,\pi\right)\cup\left[cot^{-1}\beta,\cot^{-1}\gamma\right]$, $A\in(0,\pi)$

14. The value of $|\alpha|$ is

15. The value of $\left|\alpha-60(\beta\gamma)^2\right|$ is equal to

PARAGRAPH-II

Let the complex number 'w' be one of the root of the equation

$11z^{10}+10iz^9+10iz-11=0$. Let z be a complex number satisfying $\arg\left(\frac{(z|w|)^3-1}{z^3+|w|^9}\right)=\frac{\pi}{2}$

on the argand plane(take $\pi=3.14$). On the basis of above information, answer the following questions:

16. Length of the arc of the locus of z for which $\mathrm{Re}(z) > 0$ and $\mathrm{Im}(z) < 0$ is L. then the value of 3L is ___

17. Length of the arc of the locus of z for which

PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

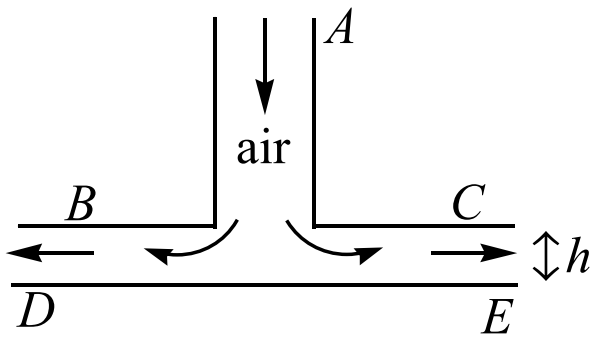
This section contains **FOUR (04)** questions.

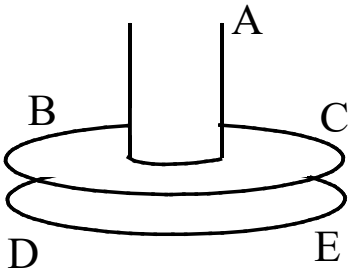
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18. DE is a flat horizontal cardboard disc of radius R over which we place another coaxial disc BC of same radius. BC is fitted with a tube A of radius r. On blowing air through A, we find that DE is attracted to disc BC. If the weight of DE is 100π gm, what should be velocity of air through A so that the attractive force can support DE? Density of air as ρ, pressure of air in A is atmospheric. R = 20 cm, r = 5 cm, h = 1 cm, $\rho = 1\ kg/m^3$. Assume that the flow is laminar.





A) 0.5 m/s approx. **B)** 13.5 m/s approx.

C) 18.5 m/s approx. **D)** None of these

19. When the mass of a critically damped single degree of freedom system is deflected from its equilibrium and released, it will

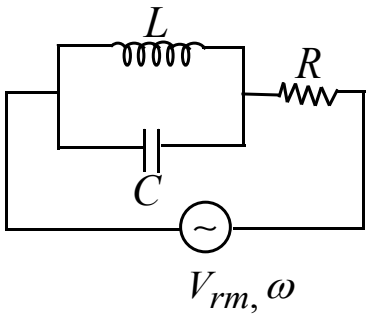
A) Return to equilibrium position without oscillation.

B) Oscillation with increasing time period.

C) Oscillation with decreasing amplitude.

D) Oscillate with constant amplitude.

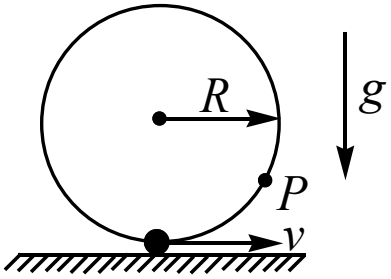
20. In the given L-C-R circuit the value of rms current through the a.c. source is



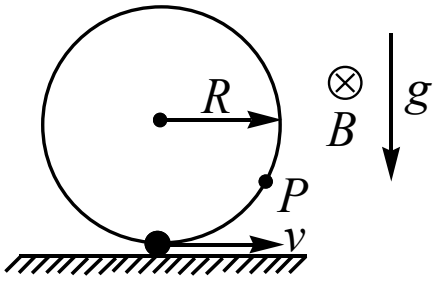
A) $\dfrac{V_{rms}}{\sqrt{R^2+\left(\omega L-\dfrac{1}{\omega C}\right)^2}}$ **B)** $\dfrac{V_{rms}}{\sqrt{R^2+\left(\dfrac{\omega L}{1-\omega^2 LC}\right)^2}}$

C) $\dfrac{V_{rms}}{\sqrt{R^2+\left(\omega^2 LC-\omega L\right)^2}}$ **D)** $\dfrac{V_{rms}}{\sqrt{(R+\omega L)^2+\dfrac{1}{\omega^2 C^2}}}$

21. In case figure (a) and (b) a positive charge particle of mass m and charge q is given velocity 'v' at the lowest point of a smooth fixed sphere of radius R. In both cases circular motion is in the plane of the paper. In case (b) magnetic field is also switched on in inward direction. If at point P, v_1 and v_2, N_1 and N_2, are velocities and normal forces in two cases respectively then.



$(a)\, v=\sqrt{2gR}$



$(b)\, v=\sqrt{2gR}$

A) $v_1 = v_2$ and $N_1 > N_2$ **B)** $v_1 < v_2$ and $N_1 > N_2$

C) $v_1 > v_2$ and $N_1 > N_2$ **D)** $v_1 = v_2$ and $N_1 = N_2$

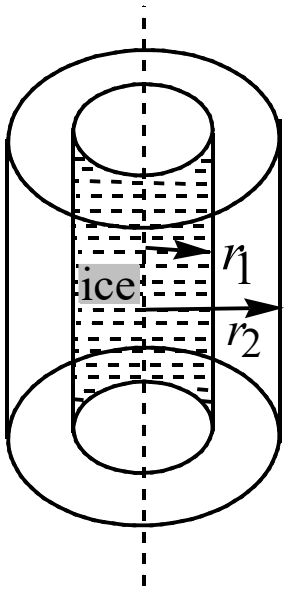
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

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22. A 100 cm long cylindrical flask with inner and outer diameter 2 cm and 4 cm respectively is completely filled with ice as shown in the figure. The constant temperature outside the flask is $40^\circ C$. (Thermal conductivity of the flask is $0.693\ W/m^\circ C, L_{ice} = 80 cal/gm$ & $\ell n2 = 0.693$).



A) Rate of heat flow from outside to the flask is $80\pi J/s$.

B) The rate at which ice melts is $\frac{\pi}{4200} kg/s$.

C) The rate at which ice melts is $100\pi\, kg/s$.

D) Rate of heat flow from outside to flask is $40\pi\, J/s$.

23. According to the second law of thermodynamics:

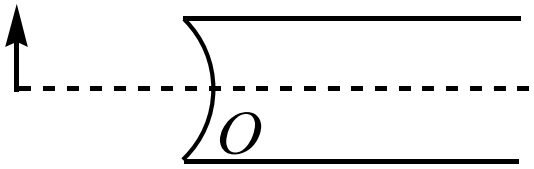
A) Heat energy cannot be completely converted to work.

B) Work cannot be completely converted to heat energy.

C) For all cyclic processes we have dQ/T < 0.

D) The reason all heat engine efficiencies are less than 100% is friction, which is unavoidable.

24. A hemisphere of radius 4 cm is carved out of a glass rod of diameter 8 cm & refractive index 1.60. An object 1 mm tall forms an erect image 4 mm tall.



A) The image must be formed in air.

B) The image is formed at a distance of 32 cm from point O.

C) The object is 1 cm away from centre of curvature of hemisphere.

D) The image must be real.

SECTION-III
(NON-NEGATIVE INTEGER)

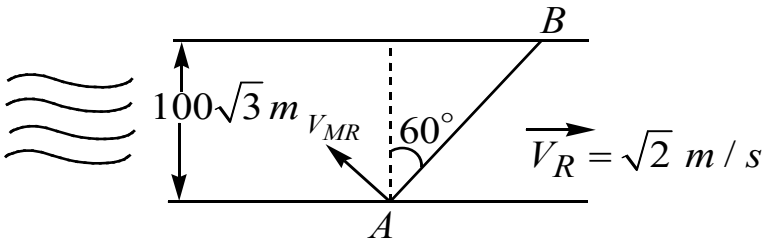
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
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Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

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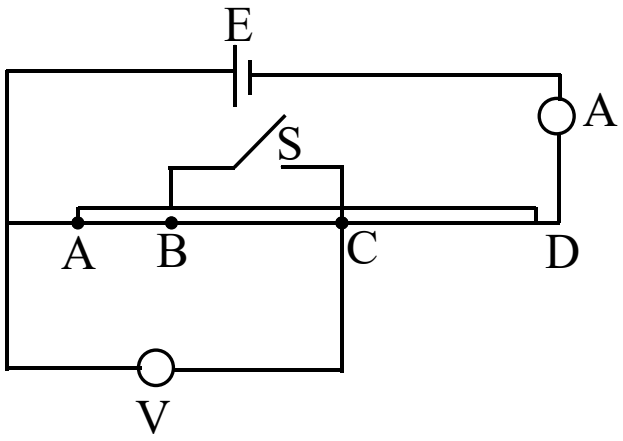
25. A man wants to reach directly at point B on the opposite bank of a river flowing at a speed as shown in fig. V_R = velocity of river with respect to ground.

V_{MR} = velocity of man with respect to river



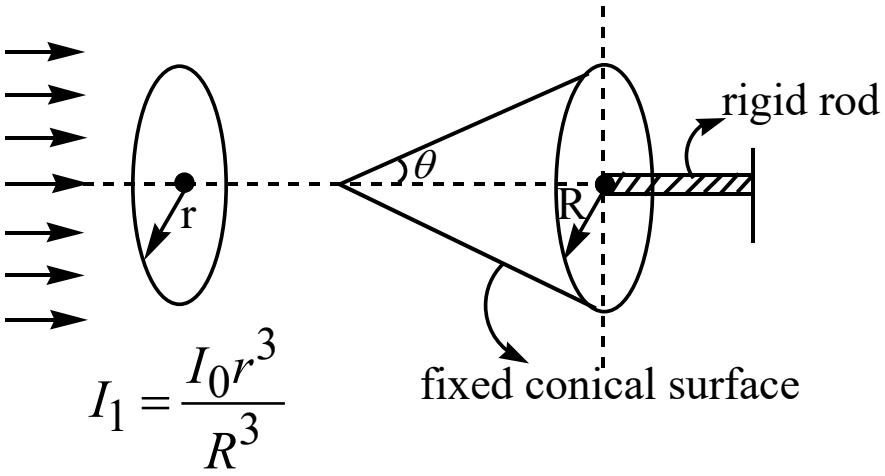
Minimum speed relative to water, the man should have so that he can reach point B will be $\frac{1}{\sqrt{k}}$ (in m/s). Find the value of k**.**

26. Figure shows a uniform wire AD of length 60 cm connected to an ideal cell, ideal voltmeter and an ideal ammeter. AB = 10 cm; BC = 30 cm. The other connecting wires are resistance-less. The reading of the voltmeter and the ammeter when switch 'S' is closed are 8V and 1A respectively. if the switch 'S' is opened then the reading of the voltmeter will be 4k volt. Find the value of k



27. In an optical bench experiment to determine the focal length of the concave mirror, the least count of the bench scale (meter scale) is 0.1 cm. The object needle reading is 0.7 cm and the image needle reading is 5.7 cm. The mirror is positioned at 15.7 cm. The maximum possible percentage error in the focal length of the mirror with this set of reading is nearly $\frac{10}{x}$. find x?

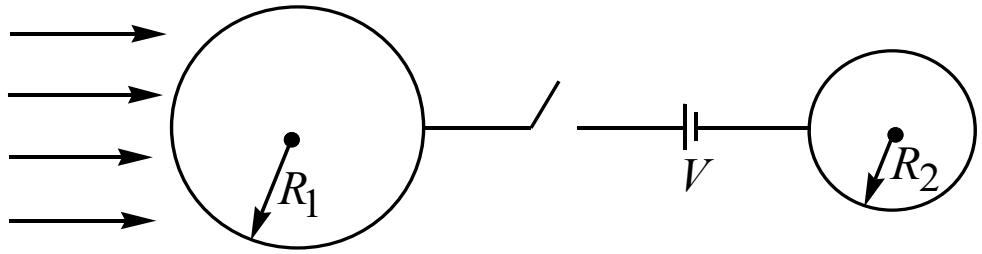
28. A cylindrical beam of light having a variable intensity I_1 is incident on the curved surface of a fixed cone of base radius R. Intensity of beam changes with radial distance r, according to the function $I_1 = \frac{I_0 r^3}{R^3}$. Conical surface is completely reflecting surface. If the force exerted by the light on the cone is of the form $\frac{k_1 \pi I_0 R^2 \sin^2\theta}{k_2 C}$ in simplest ratio. Where k_1 and k_2 are integers and C is the speed of light. Then the value of $(k_1 + k_2)$ will be?



29. Two conducting uncharged spheres of radius R_1 and $R_2 (R_1 > R_2)$ are connected to a battery with a switch as shown. Light rays of frequency f are incident on the bigger

sphere and simultaneously the switch is closed. Work function of bigger sphere is ϕ. After some time the charge on bigger sphere becomes q_1, and on smaller sphere becomes $-q_2$ remains constant thereafter.

Then $\frac{q_1}{4\pi\varepsilon_0 R_1}+\frac{q_2}{K\pi\varepsilon_0 R_2}=V$. Find the value of K



30. A body is launched from the earth's surface at an angle $\alpha=30^\circ$ to the horizontal at a speed $v_0=\sqrt{\frac{1.5GM}{R}}$. Neglecting air resistance and earth's rotation, the height from earth surface to which the body will rise is found to be K times 0.29R.

Find K. (here M is mass of earth and R the radius of earth)**.**

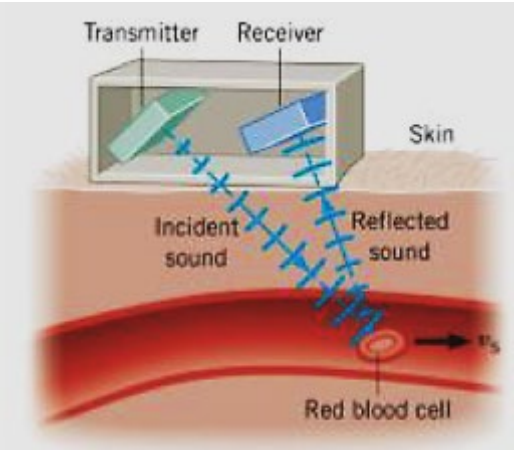
SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

The Doppler flow meter is a particularly interesting medical application of the Doppler effect. This device measures the speed of blood flow, using transmitting and receiving elements that are placed directly on the skin, as in Figure. The transmitter emits a continuous sound whose frequency is typically about 5 MHz. When the sound is reflected from the same red blood cells, its frequency is changed in a kind of Doppler effect because the cells are moving with the same velocity as the blood. The receiving element of detects the reflected sound, and an electronic counter measures its frequency, which is Doppler-shifted relative to the transmitter frequency. From the change in frequency the

speed of the blood flow can be determined typically, the change in frequency is around 600 Hz for flow speeds of about 0.1 m/s.

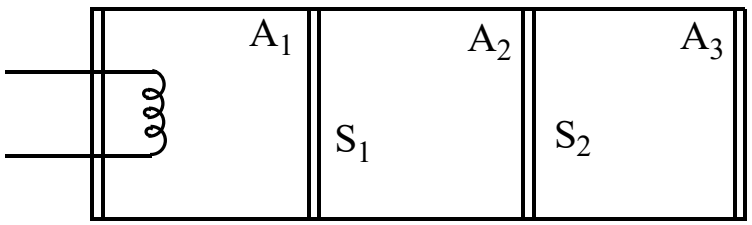


31. Assume that the red blood cell is directly moving away from the source and the receiver. The speed of the sound wave in the blood is $x \times 10^3 m/s$ .Find the value of x

32. An abnormal segment of the artery is narrowed down by an arteriosclerotic plaque to one-fourth the normal cross-sectional area. What will be the change in frequency (in Hz) due to reflection from the red blood cell in that region?

PARAGRAPH-II

Consider a closed cylinder whose walls are adiabatic. The cylinder lies in a horizontal position and is divided into three parts. A_1, A_2, A_3 by mean of a partition S_1 and S_2 which can move along the length of the cylinder without friction. Pistion S_1 is adiabatic while piston S_2 is conducting. Initially each part contains one mole of Helium gas at temperature T_0, pressure P_0, volume V_0 (see figure). Now heat is slowly supplied to the gas in part A_1 till the temperature in part A_3 becomes $T_3 = \frac{9T_0}{4}$.



33. The final equilibrium temperature of A_1 is $\frac{NT_0}{32}$ .Find N

34. The final volume of A_1 is $\frac{NV_0}{27}$.Find N

CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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35. Which of the following statements is incorrect?

A) $R_3P=O$ exist but $R_3N=O$ does not exist (R = alkyl group)

B) Among F_2, Cl_2, Br_2 and I_2; Cl_2 has highest bond dissociation enthalpy

C) Helium is used as a diluent for oxygen in modern diving apparatus because of its very low solubility in blood

D) Xenon is obtained by the interaction of XeF_4 and O_2F_2 at 143K

36. Which of the following statements is incorrect?

A) In CO_2, the carbon – carbon bond length is exactly equal to the normal carbon – oxygen 'double bond ' bond length

B) H_2CO_3 / HCO_3^- buffer system helps to maintain pH of blood between 7.28 *to* 7.42

C) Quartz, cristobalite and tridymite are the crystalline forms of silica

D) Hydrolysis of dimethyldichlorosilane followed by condensation polymerization yields straight chain polymers

37. For pure water at $50^\circ C$, ionic product of water is $10^{-13}M^2$ and density of water is 0.9 g/ml. select the correct option for H_2O at $50^\circ C$:

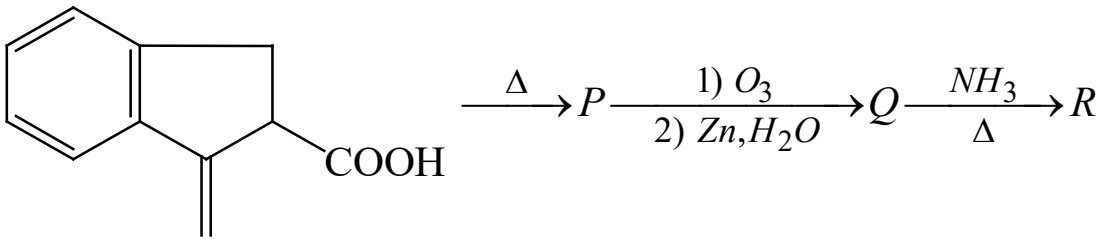
A) pH of the water is 7

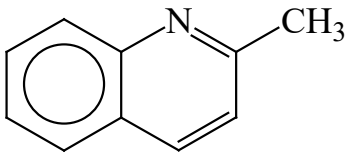
B) Dissociation constant (K_a) of the water is 2×10^{-15}

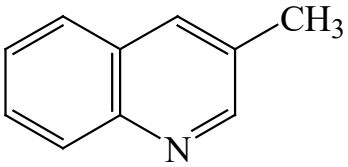
C) Degree of dissociation of the water is $2\sqrt{10}\times10^{-8}$

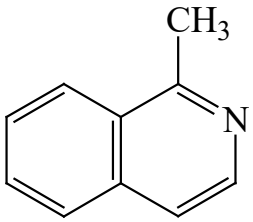
D) Any aqueous solution with pH = 6.7 is acidic

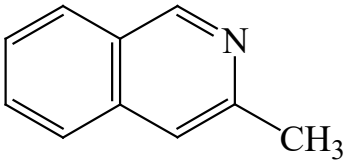
38. Major product (R) in following reaction sequence is



A)


B)


C)


D)


SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

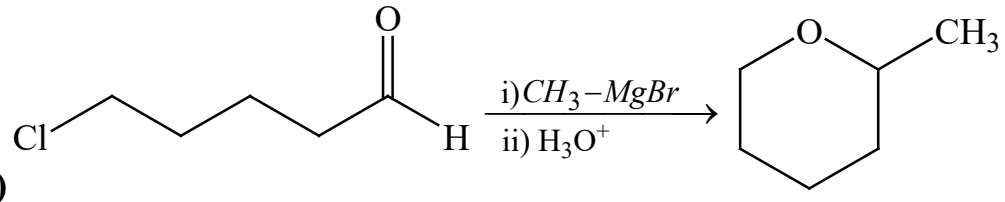
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39. The pair(s) of ions where both the ions are precipitated upon passing H_2S gas in presence of dilute HCl, is(are):

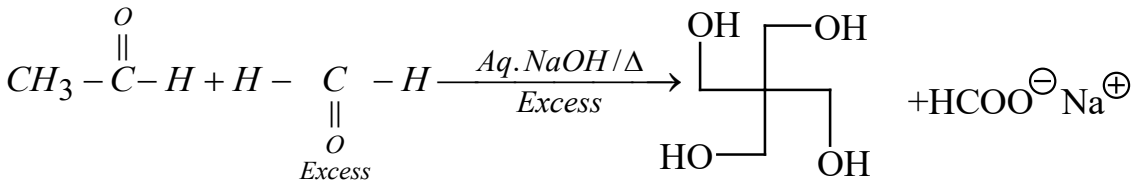
A) Hg^{2+}, Bi^{3+} **B)** Bi^{3+}, Fe^{3+} **C)** Cu^{2+}, Pb^{2+} **D)** Zn^{2+}, Co^{2+}

40. Among the following choose the reaction (s), which are correctly matched.

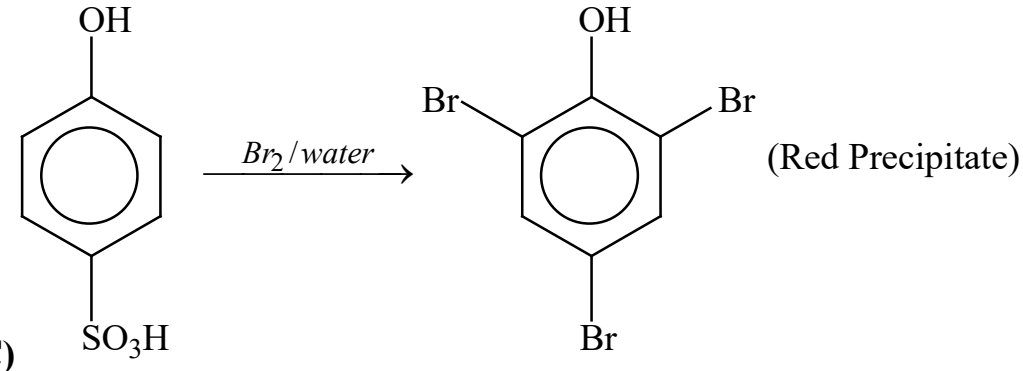
A)



B)

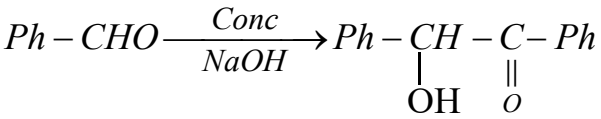


C)



(Red Precipitate)

D)



41. An ideal gas is subjected to two different reversible expansion processes, $PV^3 = K_1$ and $PV^{5/2} = K_2$ from same initial state to same final volume. which of the following statements is/are correct regarding the two processes?

A) The magnitude of work done by the gas in second process is more as compared to first process.

B) The magnitude of work done can be compared only by knowing values of K_1 and K_2.

C) Heat capacity of the gas for two processes is same.

D) final state for the two processes will be same.

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

42.

Experiment I:

During the purification of Nickel by Mond's process, if impure Nickel is heated with carbon monoxide gives a complex 'X' as intermediate compound

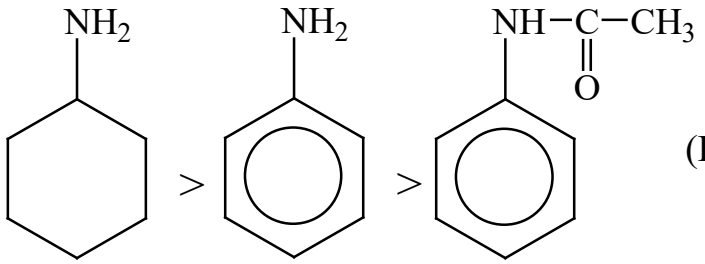
Experiment II:

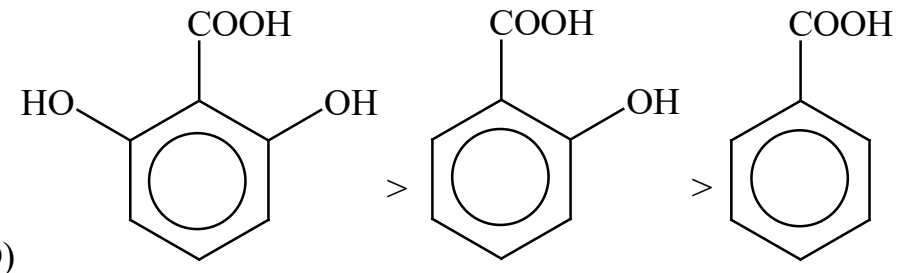
During the purification of Zirconium by Van Arkel process, if impure zirconium is heated with iodine(I_2) gives a compound Y as intermediate compound

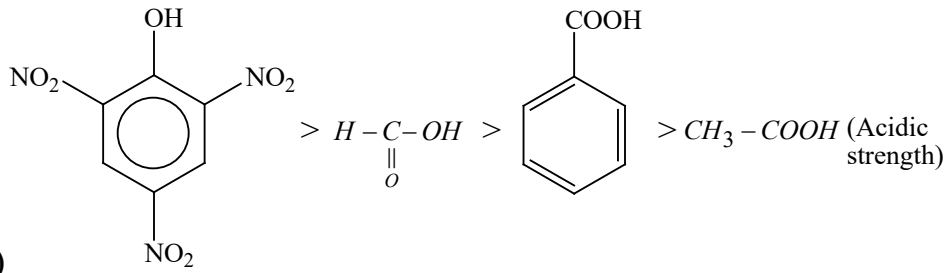
Then the total number of unpaired electrons present in X and Y is___

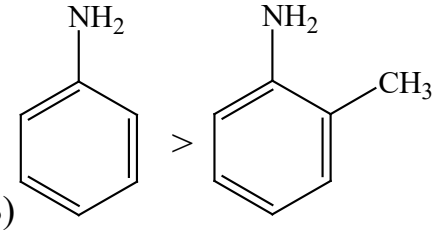
43. When ClO_2 is dissolved in alkali $(NaOH)$, two products x and y are formed. The difference in the oxidation state of chlorine in the products x and y is___

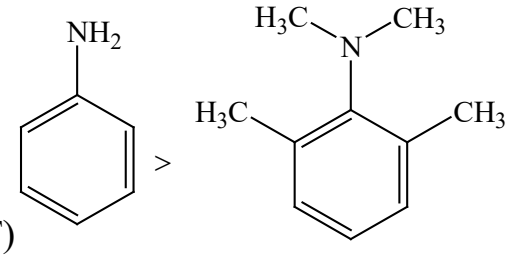
44. Some statements/orders given

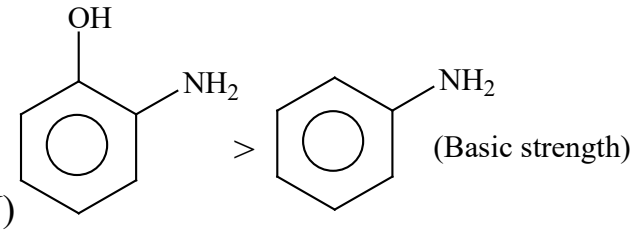
P)  (Basic strength)

Q)  (Acidic strength)

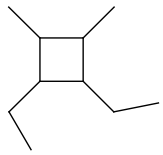
R)  (Acidic strength)

S)  (Basic strength)

T)  (Basic strength)

U)  (Basic strength)

Then the number of correct orders is/are



45. Total No of possible stereo isomers for the given compound is x.

Then find the value of x – 2

46. A student is interested in coating a metal surface of $675 cm^2$ area with 0.004 mm thickness with silver by passing a current of 96.5 amp with 50% current efficiency through a silver nitrate aqueous solution. if the time required to complete the coating is 'x' seconds then calculate the value of $\frac{x}{10}$.

(Given: Density of Ag is $10 g / cm^3$, atomic mass of Ag = 108)

47. If radiation corresponding to first line of "Balmer series" of He^+ ion is subjected to a sample of Li^{+2} ion (containing atom in different energy states) and it causes ejection of photo electron with non-zero kinetic energy then calculate least shell number in which the electron must be present in Li^{+2}

SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

PARAGRAPH-I

The rate constant of a reaction $\left[k_{forward} \right]$ increases by 7% when its temperature is raised from 300K to 310K while its equilibrium constant increases by 3%.

Given $\begin{bmatrix} \log 107 = 2.03 \\ \log 103 = 2.013 \end{bmatrix}$

48. Find out the activation energy for forward reaction in Kj.

49. Find out the activation energy for backward reaction in Kj.

PARAGRAPH-II

$Me_2CHCHO \xrightarrow[\text{ii) HBr}]{\text{i) } LiAlH_4, THF} P1$ (Major product)

iii) Sodium diethyl malonate
iv) aq.KOH
v) H_3O^+/Δ

$P1 \xrightarrow[\text{ii) } PhCH_2Cl]{\text{i) 2eq } LiN(CH(CH_3)_2)_2} P2$ (Major product)

iii) aq.NH_4Cl

50. Find the molecular weight of the product P1.

51. Find the molecular weight of the product P2.